



NUTS - THE SURPRISING HEALTH BENEFITS

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One of the most unexpected nutritional discoveries of the 1990s was that the frequent eating of nuts appears to dramatically improve health¹. In particular, nut eating greatly lowers the risk of heart disease². In 1992 researchers working on the [Adventist Health Study](#) at Loma Linda University in California reported that those eating nuts daily had up to 60% fewer heart attacks than those who ate nuts less than once per month³. The beneficial effect of nut consumption was found for men, women, vegetarians, meat-eaters, fatter people, thinner people, the old, the young, those who did much exercise and those who did little exercise. The study was large, comprising 31,000 white Californian Seventh Day Adventists and similar benefits of nut eating were subsequently found for African Americans⁴. Prior to the publication of these results, nutritional advice had usually been to minimise nut consumption on the grounds that they were a "fatty" food.

Four other large studies have since confirmed the benefits to the heart of nut eating^{2, 5-8}. In addition to the cardiac benefits of consuming nuts, the risks of having a stroke⁹, of developing type 2 diabetes¹⁰, of developing dementia¹¹, of advanced macular degeneration¹² and of gallstones¹³ have all been found to be lowered by eating nuts. Calculations suggest that daily nut eaters gain an extra five to six years of life free of coronary disease¹⁴ and that regular nut eating appears to increase longevity by about 2 years¹⁵. Adding 30 g/day of nuts to a Mediterranean diet resulted in significant reversal of the metabolic syndrome¹⁶, and consuming pistachio nuts is reported to improve erectile function in men¹⁷.

The more often nuts are eaten the better as the benefits appear to increase as the frequency of nut consumption increases. The risk of fatal coronary disease and the risk of developing type 2 diabetes both appear to decrease steadily as nut consumption increases from less than once a week to once or more per day^{10, 18}.

Just what quantity of nuts should be eaten? The studies above suggest that 30 to 60 grams (1-2oz) of nuts should be consumed daily to gain the maximum benefits seen. Whether even larger amounts confer further benefits is currently unknown.

Nuts are of course a fatty food and many might worry that they will put on weight by eating more nuts. After all, 30 grams (or one ounce) of most nuts contain about 800 kilojoules (200 kilocalories). Happily though, on present evidence, nuts do not seem to cause weight gain^{1, 19}. For example in the Nurses' Study the frequent nut consumers were actually a little thinner on average than those who almost never consumed nuts⁶, and daily supplements of almonds or peanuts for six months resulted in little or no increase in body weight²⁰. Nuts appear to satisfy hunger sufficiently well to appropriately reduce the consumption of other food.

Which nuts are best? The definitive answer to this question is currently unknown. In the Adventist Study about about 32 percent of the nuts eaten were peanuts, 29 percent almonds, 16 percent walnuts, and 23 percent other types. These researchers did not ascertain whether the nuts were fresh, oil-roasted, or dry-roasted. The Nurses' Study found that peanuts, which are legumes, appeared to be just as effective in reducing the risk of coronary heart disease as tree nuts⁶. Experiments where volunteers were fed nuts as part of their diet for several weeks have found that walnuts²¹⁻²⁴, almonds²⁴⁻²⁷, hazelnuts²⁷, peanuts²⁸, pecans³⁰, pistachio nuts³¹ and macadamia nuts³² all alter the composition of the blood in ways that would be expected to reduce the risk of coronary disease. Chestnuts, a nut unusually low in fat, do not yet seem to have been studied. The best advice currently is probably to eat a variety of nuts. Walnuts though, because they contain n-3 fatty acids, may be particularly beneficial³³, but this requires

NUTS - the surprising benefits of eating nuts regularly.

further study. Coconuts, on account of their high saturated fat content, should probably be avoided.

It is not completely understood just why nuts are so consistently found to be healthy³³. Nuts contain low levels of saturated fats and high levels of unsaturated fats. Consequently, as would be expected, studies have clearly shown that nut consumption lowers blood cholesterol levels³⁴. No doubt the lower cholesterol lowers the risk of heart disease.

Interestingly though the cholesterol is lowered by a larger amount than would be expected just from the favourable fatty acid composition of nuts³². However the reason for this is not clear. Nuts though are good sources of fibre, vitamin E, folic acid, copper, magnesium and the amino acid arginine, for each of which there is evidence of a role in preventing heart disease³³. Nuts are the best dietary source of manganese and contain plant sterols, the compounds now added to some margarines to reduce cholesterol adsorption from food, and are a good source of boron³⁵. The nutrient composition of nuts has been reviewed by King³⁶ and by Chen³⁷.

In New Zealand nuts are not a major part of the diet, most often being eaten in small quantities raw or as peanut butter. However nuts are very palatable and consumption could readily be increased. With a little imagination nuts can easily be added to many recipes including home-baked bread, cakes, soups, main dishes, sauces, stuffings, salads and desserts. Waldorf salad, pesto and baklava are internationally well known examples. Thousands of recipes using nuts are readily available on the internet, at for example [Epicurious](#), and [RecipeSource](#). In Australia and New Zealand the peanut-based meat substitutes *Nutolene* and *Nutmeat* are manufactured by the [Sanitarium](#) food company.

Nut butters too are delicious, particularly peanut, almond, cashew and hazelnut butters. However most commercial peanut butters have saturated fats added to them to prevent oil separation so that it is probably healthier to seek out brands made from peanuts (and salt) only. Unfortunately, virtually all commercial peanut butters have salt added (salt tends to raise blood pressure). Alternatively peanut and other nut butters are easily [made at home](#) by roasting, and then grinding, mincing or blending the nuts.

Nuts are an important component of the [Healthy Eating Pyramid](#)³⁸ and peanut butter was a key component in a successful weight-loss study³⁹. The success of this moderate-fat diet in enabling and maintaining weight loss has stimulated the development of a popular [Peanut Butter Diet](#)⁴⁰. Almonds were a major part of the vegetarian "Portfolio" diet⁴¹ which was just as effective for lowering cholesterol as the drug Lovastatin.

There is just one note of warning though. British⁴² and American⁴³ surveys suggest that between one in one hundred and one in two hundred people may be allergic to one or more types of nuts. Both children and adults can be affected and the most common symptoms seem to be skin rashes and hoarseness in the throat⁴⁴. For the large majority of the population though, an increase in nut consumption would appear to be desirable and "regular nut consumption can be recommended as a replacement for consumption of refined grain products or red or processed meats"¹⁰.

The Department of Nutrition of the Loma Linda University School of Public Health publishes a comprehensive online database of research articles on nuts and nutrition: <http://www.nutstudies.org/search.php>

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